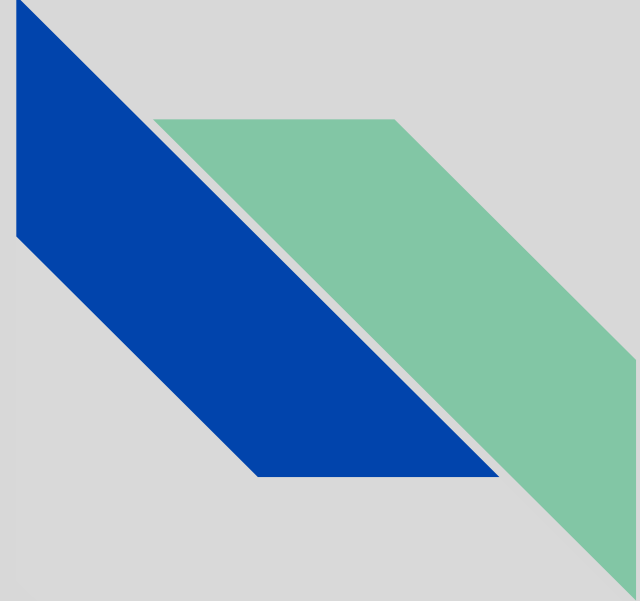




LLM RAG WITH LANGCHAIN

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Tech Start-up enthusiast (Building DEEPKA and CareMind)**

WHAT IS THE MAIN PROBLEM WHEN BUILDING LLM ?



Who was the first person to walk on the moon?

 Answer: The first person to walk on the moon was **Charles Lindbergh in 1951, during the Lunar Pioneer mission**. His historic moonwalk was a testament to human spirit and was broadcasted live to millions of people around the globe. ❌

 Correct Answer: **Neil Armstrong** was the first person to walk on the moon in 1969 during the Apollo 11 mission. ✅

(a) Factuality Hallucination



Please summarize the following news article:

Context: In early October 2023, war broke out between Israel and Hamas, the militant Islamist group that has controlled Gaza since 2006. Hamas fighters fired rockets ... civilians and taking dozens of hostages.

 Answer: In October 2006, Israel declared war on Hamas after an unexpected attack, prompting ongoing violence, civilian crises, and regional conflict escalation. ❌

(b) Faithfulness Hallucination

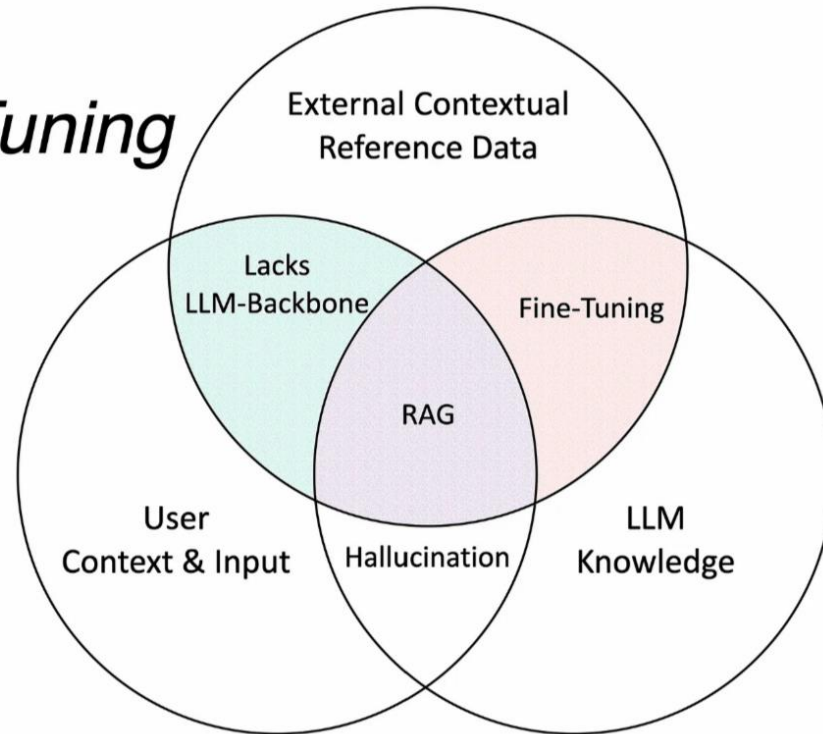
Figure 1: An intuitive example of LLM hallucination.

Teaching LLMs To Say,
“I don’t know”

Based on a recent research paper.

WHAT IS THE SOLUTIONS?

RAG & Fine -Tuning





WHAT IS THE SOLUTIONS?

Hallucination is when the LLM provides highly plausible, concise, and seemingly credible responses that are factually inaccurate.

Fine-Tuning changes the models behaviour while RAG provides external contextual data to reference at inference.



HOW TO DO RAG ?

Phase 1—Setup Memory for LLM (Vector Database)

- Load raw PDF(s)
- Create Chunks
- Create Vector Embeddings
- Store embeddings in FAISS

Phase 2—Connect Memory with LLM

- Setup LLM (Mistral with HuggingFace)
- Connect LLM with FAISS
- Create chain

Phase 3—Setup UI for the Chatbot

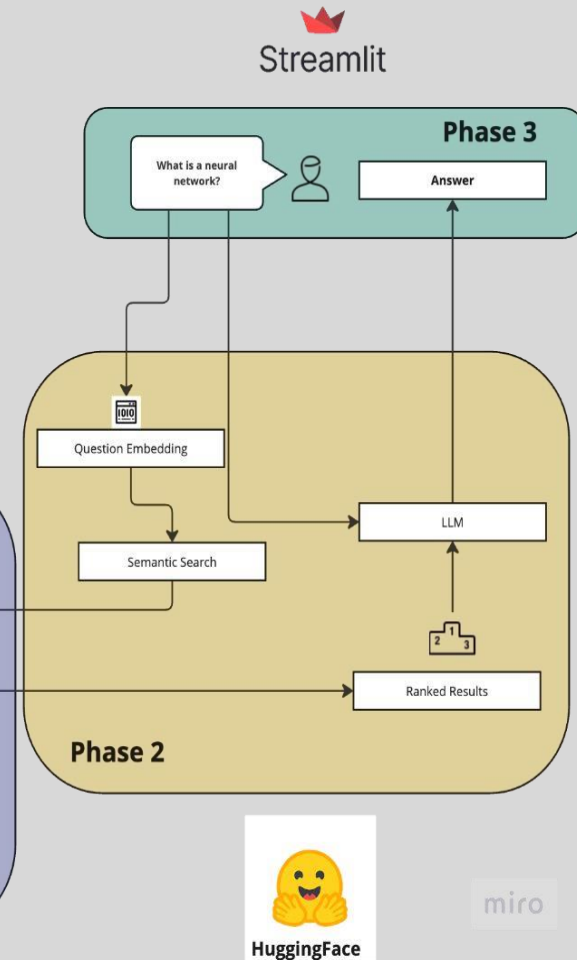
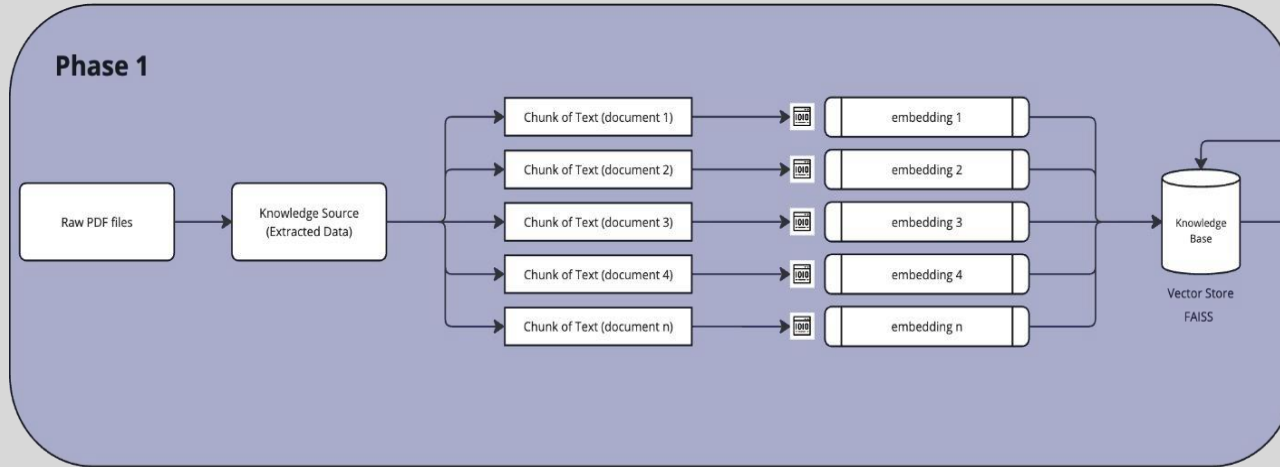
- Chatbot with Streamlit
- Load Vector store (FAISS) in cache
- Retrieval Augmented Generation—RAG



Tools & Technologies

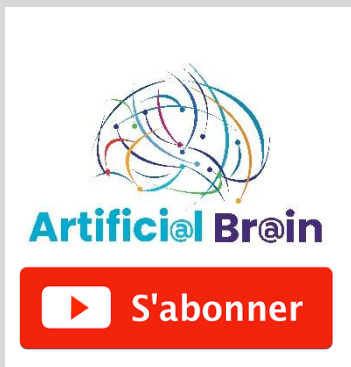
- Langchain (AI Framework for LLM applications)
- HuggingFace (ML/AI Hub)
- Mistral (LLM Model)
- FAISS (Vector Database)
- Steamlit (For Chatbot UI)
- Python (Programming Language)

Technical Architecture





CASE STUDY: Doctor CareMind



THANKS ! Questions ?

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